ECHO-7

Integrated vision-guided bay search and docking

DATASHEET / 05 NOV 2026

Design E7070008 / CCTL E7070001

SYSTEM / CURRENT IMPLEMENTATION

DE1-SoC + OV7670 + UGV02 | Cyclone V 5CSEMA5F31C6 | Production top: **echo7IntegratedTop**. One SOF selects real vision, synthetic-vision chassis debugging or integrated operation. Physical IR and J3 sonar always remain in the motion safety chain.

MODE	INPUT / OUTPUT
INTEGRATED (power-up default)	Real camera; explicitly started tasks only. No host heartbeat required.
VISION_ONLY	Real tracking/calibration; wheel UART remains zero, even without a valid sonar reading.
CHASSIS_INJECT	JTAG observations at 25 Hz through the real C controller; physical IR/J3 remain active.

CONTROLS & VISUAL WORKFLOW

SW0-SW5: one-hot bay 1-6; approximately 20 ms stability. No selection/conflict cannot move. Selection changes cancel the task; valid selection tracks before ARM.

SW6/SW7: candidate / centre RGB-HSV diagnostics. All 16 candidates appear on two 8-slot pages, about 2 s/page. **SW8:** perspective bypass. **SW9:** raw display, cancel motion.

Red1: colour calibration. **Red2:** reselect switches and request a task. Uncalibrated mode waits for a new 3-frame lock. Three healthy frames classified absent, then three present, retrigger; continuously recognized presence does not re-arm.

IR A (0x0F): latched stop. **POWER (0x12):** first fresh press releases an IR stop; a second press starts. Repeats/held POWER do not start twice. **KEY0:** full reset.

MOTION & STOP CONDITIONS

Search: opposite logical wheel requests at 4 cm/s. First valid observation stops rotation; await a complete zero-speed JSON, then 1.5 s conservative dwell and 3 new independent frames.

Alignment: U=110-130 centred; bounded 3/4 cm/s turns for 20-100 ms, then reacquire. **Approach:** centred 5/5 cm/s above 500 mm, otherwise 3/3. Within U75-165, drift correction uses 5/3 or 4/3. Outside this range or at 350 mm off-centre: stop, dwell and reconfirm before pivoting. **Dock:** at 350 mm only when centred and locked.

Guard: distance <=250 mm, faulty or stale data stops motion. A completed out-of-range echo before the 80 ms deadline is **FAR**, allowing travel/search; the d page shows **FAr**. HOLD recovery needs 3 new safe samples >=300 mm. DOCKED/NOT_FOUND require a new start.

Freshness: visual reports about 150 ms; injection lease 500 ms (host renewal about 100 ms). Lost lease cancels ARM; renewed heartbeat does not restart. UART: 115200 8N1, complete JSON, zero-speed priority.

CALIBRATED_TIME 13.6 s restored on the unchanged plastic floor using three E5 circles (11.075500-13.561176 s; range 2.485676 s). SEARCH speed remains 4 cm/s. Bounded approximate scan only, with overshoot/drift and no yaw feedback. Epoxy untested; fixed-budget angle unmeasured. Reset/reprogramming requires restoring the runtime calibration.

EVIDENCE / CURRENT BUILD & HARDWARE

CHECK	RESULT / LIMIT
Directed control & protocols	PASS: both steering signs, 433 checks / 88 complete UART messages each. Forward 5/3 and 4/3 corrections, boundary stops, zero/dwell/fresh qualification and safety covered.
Pixel-to-UART system	PASS: 14 fixed-rate 25 fps frames through perceptionCore, real C and UART. Synthetic pixels; shortened test dwell, not live-camera or physical-settling acceptance.
Course R6	PASS 270.641 s, E8; clean Verilator 5.050, jobs=4; 13 PASS + mutant CAUGHT; below 285 s. Fourteen 25 fps pixel frames include actual 3/5 forward-correction UART.
Quartus fit / timing	PASS: compile 1,159.722 s + gate 25.963 s. ALM 29,281/32,070 (91%); M10K 357/397; DSP 51/87; 41,081 registers. Worst setup +1.021 ns / hold +0.121 ns; unconstrained 0; 0 errors / 92 warnings.
Board / physical checks	E8 blue1 real-camera task reached DOCKED in 29.315 s, then >3 s disarmed/zero. User: continuous forward travel, no obvious turn, ruler ~335 mm; J3 329 mm. Final fine alignment took ~25 s; possible ~1 cm unequal wheel travel remains. This trial began visible with one host START; five other bays and search/red2/POWER tasks remain pending.
Earlier wheel / IR checks	E4 raised: real-J3 slowdown, docking, hysteresis/recovery and persistent STOP confirmed; earlier real IR A/POWER and both wheels forward. E5/E7 failed real ground tasks remain documented.

Remaining: Five further real-camera bay tasks, search/red2/POWER integration; slow final alignment, startup asymmetry, fixed-budget angular error, scene-to-UART latency and SW8 ground comparison. A/C student IDs pending.

Current SOF: firmware/E7070008/PartBDebug.sof; 6,706,156 bytes. Original build: 20261104T160829_127927Z_build. Programmed-board ID E7070008 verified.
Reproduction: README.md / README_EN.md; docs/REPAIR_RESULT.md; tools/render_integrated_datasheet.py. No direct wheel or synthetic-distance override.